

Receiving Help and Short-Run Contribution on Stack Overflow: A Newcomer-Concentrated Association

Abstract

Whether receiving help on an online platform increases subsequent helping is an open empirical question. Prior observational studies report mixed results, and designs that do not net out baseline user activity may overstate reciprocity. We develop a matched difference-in-differences survival analysis of over 21 million Stack Overflow questions, comparing users whose questions received an answer with propensity-score-matched users whose questions did not. Receiving an answer is associated with a small increase in subsequent helping (pooled summed DiD HR = 1.03), but the association is steeply graded by tenure: it is largest among newcomers (HR = 1.13 in the first week) and undetectable among users with more than three years of experience. The newcomer gradient is the most robust pattern in the data — it holds on the pre-2020, pre-generative-AI subsample — yet the pooled effect is modest in absolute terms and reverses once we control for the quality of the answer received, and answer speed moderates it weakly and inconsistently. The data identify short-run behavioral associations, not the psychological mechanisms (gratitude, moral obligation, incentive displacement) that theory proposes. These findings provide a better-identified short-run behavioral test of generalized reciprocity in one large technical Q&A community and suggest tentative, newcomer-focused design implications.

Keywords: generalized reciprocity, online communities, knowledge sharing, survival analysis, platform design, Stack Overflow

1. Introduction

Whether receiving help on an online platform motivates subsequent helping behavior has been a persistent question in the study of online communities. Generalized reciprocity, the tendency to help others after being helped oneself (Nowak and Roch, 2007; Baker and Bulkley, 2014), is among the

mechanisms proposed to sustain cooperation on platforms like Stack Overflow and Wikipedia, where direct exchange between the same two individuals is rare. If receiving help reliably increases the probability of giving it, cooperation may sustain itself through cascading helping acts, even among strangers (Fowler and Christakis, 2010). The behavioral evidence, however, is inconsistent: some observational studies find substantial reciprocity effects (Baker and Bulkley, 2014; Chen et al., 2019), while others find weak or null relationships (Wasko and Faraj, 2005; Joyce and Kraut, 2006). We argue that much of this inconsistency is methodological. Our contribution is a better-identified short-run behavioral test on one platform: whether helping rises after answer receipt. The data do not directly measure the psychological mechanisms that theory proposes (gratitude, moral obligation, incentive displacement); we test only the behavioral consequence those mechanisms predict.

Observational studies of reciprocity on platforms face a persistent confound: more engaged users are more likely both to seek help and to provide it, regardless of whether their own questions are resolved. Studies that correlate help received with help given, without accounting for this baseline activity, conflate reciprocity with general user engagement (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). A related limitation is that prior designs have largely treated receiving help as a one-time event, even though help-seeking and helping occur continuously.

We address both problems by developing a matched difference-in-differences survival analysis, applied to over 21 million questions from Stack Overflow. For each question a user posts, we define observation windows before and after the question and compare changes in helping behavior between users whose question received an answer (treated) and propensity-score-matched users whose question did not (control). We use Cox proportional hazards models to account for the continuous-time dynamics of helping behavior and the staggered, time-varying structure of treatment.

We develop two theoretical predictions alongside the basic reciprocity hypothesis. Drawing on theories of socialization and norm internalization, we predict that the reciprocity effect is stronger among newcomers than among experienced users (H2). Newcomers have not yet internalized community-specific norms or learned to value platform incentives such as reputation points; general moral obligations to reciprocate are therefore more likely to drive their behavior. As users accumulate experience, these general-purpose triggers may be displaced by community-specific motivations. We also pre-

dict that faster responses should amplify the reciprocity effect (H3), because gratitude is more intense when help arrives while the need is still salient (McCullough et al., 2001; Tsang, 2006).

The results offer qualified support for H1 and H2 and little support for H3. Relative to matched controls and to their own pre-question baseline, users who receive an answer show a small but statistically clear elevation in subsequent helping (a composite of answers and comments posted to other users; self-directed accepts excluded): the pooled summed difference-in-differences hazard ratio is 1.03 (95% CI [1.02, 1.04], $p < .001$). This pooled effect is modest in absolute terms and, as we show, does not survive conditioning on answer quality. The elevation is steeply graded by tenure: it is largest among newcomers (HR = 1.13 in the first week, $p < .001$) and indistinguishable from null among users with more than three years on the platform. We report the treatment effect as the summed post- versus pre-question contrast, decompose it into waiting-period and answer-arrival components, and separate answers from comments (Table 6) to locate the signal.

Our study makes three contributions. First, using a design built to remove the activity confound that inflates observational estimates, we provide a better-identified short-run behavioral test: receiving an answer is associated with a small increase in subsequent helping on Stack Overflow (pooled HR = 1.03), an association that falls below one once answer-quality controls are added and is therefore best read as an upper bound. Second, we show that this association is steeply concentrated among newcomers and absent among veterans — the tenure gradient is the most robust and informative pattern in the data, though we cannot cleanly separate reciprocity from selection into being answered. Third, methodologically, we decompose the outcome by help type and enter the waiting period as an explicit term, making pre-answer divergence visible rather than silently absorbed into the treatment effect.

2. Theory

2.1. Generalized Reciprocity

A central mechanism proposed to support cooperation in online communities is *generalized reciprocity*, the tendency for individuals who have received help to help others in turn, even when the recipient is not the original benefactor (Nowak and Roch, 2007; Fowler and Christakis, 2010). Unlike direct reciprocity, which involves returning favors to the same individual,

generalized reciprocity requires no repeated interactions or exchange with the same person. Instead, it operates as a pay-it-forward model, sustaining collective contributions within groups where members may have infrequent or non-recurring interactions, such as online knowledge-sharing platforms.

Socio-psychological research identifies two complementary mechanisms underlying generalized reciprocity: moral reasoning and emotional responses. On the moral side, individuals may be guided by the logic of universalization (Levine et al., 2020): they consider what would happen if all members requested more help than they gave. Such deliberation typically leads individuals to conclude that widespread free-riding would undermine the community as a valuable source of help, generating a shared moral obligation to reciprocate (Curry et al., 2019). This renders the relationship between individual and group into an indirect exchange: if one receives help from the group, one feels obligated to help the group in return.

Emotional mechanisms reinforce these moral considerations. A prominent response is gratitude, aroused by the perceived generosity of others (McCullough et al., 2001, 2008). Gratitude has been shown to motivate prosocial actions toward others beyond direct reciprocation (Tsang, 2006; Bartlett and DeSteno, 2006). According to Molm et al. (2007), such emotional responses are especially relevant for systems where free riding is easy, such as online knowledge-sharing platforms. Experiencing gratitude in these contexts not only increases the propensity to help but also strengthens perceptions of the community as valuable and worth sustaining (Elster, 2009). These moral and emotional pathways are the *proposed* generative mechanisms behind generalized reciprocity; our design tests only whether helping rises after answer receipt, not whether gratitude or moral obligation is experienced.

Generalized reciprocity is among the most widely studied behavioral mechanisms explaining cooperative behavior in groups, owing to its potential to initiate cascading acts of generosity (Fowler and Christakis, 2010; Boyd and Richerson, 1989). Comparative research and meta-analyses indicate that it is a globally shared moral principle (Curry et al., 2019). Robust experimental evidence comes from both laboratory (Greiner and Levati, 2005; Molm et al., 2007; Stanca, 2009) and field settings (Mujcic and Leibbrandt, 2018; Fowler and Christakis, 2010).

2.2. Empirical Challenges in Measuring Generalized Reciprocity

Despite its theoretical importance, empirical findings on generalized reciprocity in online communities are mixed, and we argue that this inconsistency

is largely attributable to persistent methodological challenges. Understanding these challenges is essential both for interpreting existing findings and for motivating our analytical approach.

A first challenge concerns the reliance on self-reported motives in surveys and interviews. Contributors frequently describe reciprocity as a central reason for their engagement (Wasko and Faraj, 2000; Lakhani, 2003), yet sociological and psychological research has demonstrated that people have difficulty articulating the moral motives behind their actions (Vaisey, 2009; McClelland et al., 1989). When users state that they help because they enjoy it or feel a sense of obligation, it is unclear whether this reflects genuine reciprocity, altruism, conformity to perceived social norms, or the popular rhetoric of ‘giving back’ as a post-hoc rationalization. The distinction between reciprocity and other prosocial motives, such as the intrinsic joy of helping or the pursuit of social status, is particularly difficult for individuals to make through introspection (Batson, 1991; Bénabou and Tirole, 2006). As a result, self-reported data might overstate the prevalence of reciprocity as a discrete motive.

A second and more fundamental challenge is the endogeneity of help-seeking and help-giving in observational data. Users who ask questions on platforms like Stack Overflow are also disproportionately likely to answer questions, not because receiving help triggered a reciprocal response, but because both behaviors reflect the same underlying characteristics, such as greater platform engagement. Studies that correlate help received with help given without accounting for this baseline activity effectively confound reciprocity with user type (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). As we discuss below, the confound between reciprocity and user type is the central identification challenge that motivates our matched diff-in-diff design.

Prior observational studies of online reciprocity report mixed magnitudes. Baker and Bulkley (2014) and Chen et al. (2019) find positive associations between help received and subsequent helping; Wasko and Faraj (2005) and Joyce and Kraut (2006) find weak or null relationships. Our matched DiD estimates sit below the larger positive reports (pooled summed DiD HR ≈ 1.03 ; newcomer HR ≈ 1.13), consistent with the view that designs that do not net out baseline activity overstate the effect. The tenure gradient — clear elevation among newcomers, null among veterans — offers one reconciliation of the mixed prior findings.

2.3. Reciprocity and the User Trajectory

We propose that the mixed findings in the literature can be partly reconciled by attending to a dimension that prior research has largely overlooked: the user trajectory from newcomer to experienced contributor. We argue that generalized reciprocity is strongest for new users and attenuates as users gain experience on the platform.

The logic of this prediction draws on theories of socialization and norm internalization. When a user first encounters an online community, they are unfamiliar with its specific norms, social structures, and reward systems. At this stage, community-specific incentives, such as reputation points, badges, or leaderboard positions, carry little motivational weight because the newcomer has not yet learned to value them. Instead, initial contributions are guided by more general moral principles that operate independently of community-specific knowledge. Generalized reciprocity is precisely such a principle: it requires only that the individual has received help and feels a moral obligation or emotional impulse to help in return, without any need to understand or identify with the particular community (Levine et al., 2020; Curry et al., 2019).

As users interact with the community, they gradually learn its norms and develop a sense of membership. On platforms like Stack Overflow, these norms are codified in explicit reputation systems that reward helpful behavior with visible status markers. Once users internalize these community-specific norms, the broad moral obligation of reciprocity may be displaced by more localized incentive structures — a theoretical prediction our design cannot adjudicate directly, because we do not observe reputation salience or user motives. The logic of universalization (“I should help because I was helped”) gives way to the logic of community membership (“I should help because that is what valued members do”). This norm displacement has been theorized in related contexts by Bénabou and Tirole (2006), who show how external incentives can crowd out moral motives.

Several additional mechanisms accelerate this decline. First, as users accumulate experiences of receiving help, each additional instance becomes less socially surprising and less likely to elicit strong gratitude (McCullough et al., 2008). The emotional response that fuels reciprocity habituates over time. Second, as users observe the platform’s incentive structure that rewards helpful behavior with reputation and status, they may increasingly attribute others’ helpful behavior to status pursuit rather than genuine generosity, further dampening the gratitude that triggers reciprocal helping.

This account predicts that generalized reciprocity is strongest among newcomers and declines with experience. This also helps explain why prior studies have reached different conclusions about the importance of reciprocity: those that focus on or oversample experienced users may underestimate reciprocity, while those that capture newcomer behavior may find stronger effects.

HYPOTHESIS 1 (H1). *Receiving help increases a user’s propensity to help others (generalized reciprocity effect).*

HYPOTHESIS 2 (H2). *The generalized reciprocity effect is strongest among newcomers and decreases as users gain experience on the platform.*

2.4. Response Time and the Immediacy of Reciprocity

The emotional mechanisms underlying reciprocity suggest that the speed of help matters for the strength of reciprocity. Gratitude, the primary emotional driver of generalized reciprocity (McCullough et al., 2001), is an affective response that is triggered by the perception of a benefactor’s generosity. The intensity of this emotional response is likely to depend on the characteristics of the helping experience, including its timeliness.

A swift response to a help request can amplify the experience of receiving help through several channels. First, a fast answer signals that the community is attentive and generous, strengthening the perception that the group is worth sustaining through reciprocal contributions. Second, gratitude is an emotional response that may be most intense when it is temporally proximate to the perceived need. When a user posts a question, their need for help is salient; a quick answer arrives during this window of heightened need. The longer the user waits for help, the less valuable the help is and the more likely they are to have gotten help elsewhere. Thus, quick responses are likely to intensify feelings of gratitude.

Conversely, delayed responses may attenuate reciprocity. When an answer arrives hours or days after the question was posted, the user’s original need may have faded; they may have found a solution elsewhere, or the problem may have become less pressing. The interval between asking and receiving weakens the associative link between the two events, potentially diluting the emotional response. Furthermore, longer waits may signal that the community is less responsive or generous, weakening the perception that underpins the moral obligation to reciprocate.

These considerations lead to a straightforward prediction:

HYPOTHESIS 3 (H3). *Faster responses to a user’s question strengthen the generalized reciprocity effect, while slower responses weaken it.*

We note that this hypothesis follows from the dominant theoretical accounts of generalized reciprocity, which emphasize emotional immediacy and the salience of the helping experience. However, as we will show, field settings introduce structural dynamics, particularly the behavioral consequences of users being online and offline, that qualify the predictions of laboratory-based theory. We return to this issue when presenting our results.

3. Empirical Setting: Stack Overflow

Stack Overflow offers a valuable empirical setting for examining generalized reciprocity in online knowledge sharing. The platform provides publicly available quarterly data dumps containing rich longitudinal information on user behavior, including knowledge-seeking and knowledge-providing activities dating back to the platform’s inception. This extensive dataset enables the reconstruction of helping patterns over time, allowing for the analysis of temporal sequences of receiving and providing help, which is critical for studying generalized reciprocity. Stack Overflow’s active question-and-answer volume has declined sharply since the release of generative AI tools in late 2022; our sample spans platform inception through 1 April 2025 and therefore captures a period that includes but predates the steepest LLM-driven decline. We treat the findings as a historical snapshot of one large, technical, relatively anonymous, status-driven platform; external validity to other communities is discussed in Section 6.

At its core, Stack Overflow is a question-and-answer platform for software developers and technologists. As of mid-2025, the site had accumulated over 24 million questions and 36 million answers from more than 4.5 million users. Its user base spans a wide range of experience levels and professional roles in software development and learners (Stack Overflow, 2024). Questions on the platform must be tightly focused on specific programming problems; broad or opinion-based inquiries are typically closed by community moderation (Stack Exchange Inc., 2024). The most frequently discussed topics, as reflected in question tags, include JavaScript, Python, Java, C#, PHP, Android, and HTML, spanning areas from web development and mobile applications to

data science and machine learning. Stack Overflow is built around several core features: tags categorize questions by topic, facilitating expertise matching; a voting system allows users to upvote or downvote questions and answers to assess quality; and reputation points are earned for valued contributions such as providing high-quality answers; (Stack Exchange Inc., 2024).

For our study, the key feature is the question-and-answer interaction between users. When a user posts a question, other users may provide answers. This creates a clearly timestamped sequence (question posted, answer received) that we exploit to measure the temporal dynamics of generalized reciprocity. Crucially, on Stack Overflow, the identities of question-askers and answerers are typically different individuals, making this an ideal context for studying *generalized* reciprocity: users who receive help from one community member may subsequently help entirely different members.

4. Data and Methods

To test our predictions, we develop a matched staggered difference-in-differences (DiD) survival analysis that leverages the temporal structure of help-seeking and help-giving on Stack Overflow. Our design compares changes in helping behavior (measured as the hazard of posting an answer) before and after an user receives an answer, between users who receive an answer and matched users who do not. This approach represents a novel integration of two methodological traditions: the staggered difference-in-differences framework of Callaway and Sant’Anna (2021), which accommodates variation in treatment timing and treatment effect heterogeneity, and the counting-process formulation of the Cox proportional hazards model (Andersen and Gill, 1982), which allows treatment status and other covariates to vary continuously over the observation window.

4.1. Study Design

For each question posted by a user, we construct a timeline centered on two key events: the posting of the question (T_{question}) and, for questions that receive an answer, the time at which the first answer with a non-negative vote score is posted (T_{answer}). We define an observation window spanning the two days before the question is posted ($T_{\text{question}} - 2\text{D}$) through the two days after ($T_{\text{question}} + 2\text{D}$). This yields a 4-day timeline for each question, divided into distinct phases.

We define three phases within each observation window (see Figure 1). The first phase, the *pre-question phase*, is from the start of the observation window to the posting of the question ($T_{\text{question}} - 2D$ to T_{question}). This phase captures baseline helping activity before the focal question is asked. The second phase, the *waiting period* is from the posting of the question to the receipt of the first answer (T_{question} to T_{answer}). During this interval, the user has asked for help but has not yet received it. The final *post-answer phase* is from the receipt of the answer to the end of the observation window (T_{answer} to $T_{\text{question}} + 2D$). This is the phase in which the reciprocity effect, if it exists, should manifest.

A key feature of our design is that the post-answer (treated) phase begins at different elapsed times across observations, depending on how quickly the posted question receives a response.

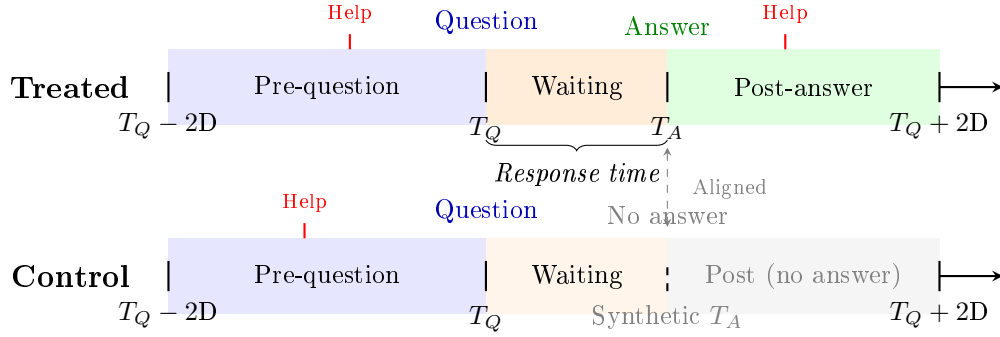
This staggered timing of the post-answer phase corresponds to the setting studied by Callaway and Sant’Anna (2021), who show that standard two-way fixed-effects estimators can be biased when treatment timing varies, and treatment effects are heterogeneous. Our approach avoids this pitfall by explicitly modeling the time-varying change in treatment status within each observation window and by comparing treated observations only to not-yet-treated or never-treated controls.

For control users (those whose questions did not receive an answer), there is no natural transition between the waiting period and the post-answer phase. To maintain temporal alignment with the matched treated observation, we assign the control observation a synthetic phase transition at the same elapsed time as the treated user’s response time. That is, if the treated user’s answer was posted 4 hours after their question, the matched control observation’s timeline is split at $T_{\text{question}} + 4h$, creating an aligned “post” phase of equal duration. This ensures that any differences between treated and control observations in the post-answer phase are not driven by differences in the length of the observation period.

The survival clock in our Cox models is centered on posting of the respective question: time zero corresponds to the start of the observation window ($T_{\text{question}} - 2D$), so that the question itself is posted at day 2 of the timeline and the observation window closes at day 4. All help events (answers posted by the focal user) are recorded with their exact timestamps and converted to elapsed time on this clock.

Within each timeline, we track all instances in which the user helps another user — by answering their question or by commenting on their content.

Figure 1: Study Design: Matched Difference-in-Differences Survival Analysis



Note. Each observation window spans 4 days centered on the focal question (T_Q). The treated timeline (top) shows three phases: the pre-question phase (baseline), the waiting period (question posted, no answer yet), and the post-answer phase (after the first qualifying answer T_A arrives). The control timeline (bottom) is from a propensity-score-matched user whose question did not receive an answer. To ensure temporal alignment, the control's post-question period is split at a synthetic transition point matching the treated user's response time. Red markers indicate help events (answers posted by the focal user to other users' questions). The treatment effect is the differential change in helping hazard between treated and control observations in the post-answer phase, relative to the pre-question baseline. The staggered timing of T_A across observations motivates our use of matched DiD.

These helping events constitute the outcome in our survival analysis. Conceptually, our design asks: Does the hazard of helping increase after a user receives an answer, relative to the change in hazard for a matched user who does not receive an answer? We present descriptive evidence on the raw help rates over this timeline in Section 5 (Figure 2), before turning to the formal Cox model estimates.

4.2. Data

We utilized the full sample of non-deleted questions from Stack Overflow’s inception on 31 July 2008 through 1 April 2025. We addressed two edge cases that would render our fixed-length observation window invalid. First, for early platform questions, the pre-period could predate the platform’s launch, meaning no answers were possible at that time. We removed 1,080 such cases. Second, due to the data dump’s April 1, 2025, cut-off, the post-period could be prematurely truncated. We excluded 8,029 cases affected by this. We then excluded 2,671,121 questions that received an answer from the question poster. These filtering steps resulted in a pre-matching dataset of 21,033,429 questions from 4,764,048 unique users. After propensity score matching (described below), the matched file comprises 24,527,534 question-level rows (12,263,767 matched pairs; some control questions appear more than once) from 3,899,114 unique users, with a 50% treatment rate by construction. All matched rows enter the Cox analysis; the helping outcome is a composite of answers and comments the focal user posts to other users’ content (self-directed accepts excluded), yielding 640,237 helping events in the pooled model (Table 5).

4.3. Variables

Table 1 provides an overview of the variables used in our analysis. The phase terminology is consistent with Figure 1: the pre-question phase captures baseline behavior, the waiting period spans the interval from question to answer, and the post-answer phase captures the period after the answer arrives. We use “post-question” to refer to the combined waiting period and post-answer phase whenever distinguishing between these two sub-periods is not necessary.

Outcome. The event of interest is a *helping act* directed at another user. Our default outcome is a composite that counts, within the observation window, both answers the focal user posts to other users’ questions and comments

Table 1: Overview of Variables

Variable	Description
<i>Outcome (Survival Event)</i>	
helpEvent	A binary event indicator equal to 1 at times when the user posts an answer or a comment on another user’s content within the observation window (self-directed accepts excluded).
<i>Time-Varying Covariates</i>	
PostQuestion	Binary; 1 for intervals occurring after the user posts their question ($t \geq T_{\text{question}}$), 0 otherwise. Captures the combined post-question period (waiting period + post-answer phase).
PostAnswer	Binary; 1 for intervals occurring after the user receives their first answer ($t \geq T_{\text{answer}}$), 0 otherwise. For control users (no answer received), this variable switches on at the synthetic transition point at the matched treated user’s response time. Isolates the post-answer phase.
treatedPostQuestion	Interaction of treatment $\times$ phasePostQuestion; captures differential behavior of treated users in the entire period after the question.
isTreatedActive	Interaction of treatment $\times$ phasePostAnswer; the primary treatment effect capturing the additional lift in helping hazard after receiving an answer, beyond any change attributable to the act of asking a question.
ResponseTime	Time between question and the first answer (excluding down-voted answers) of the treated question. We include interactions with all four variables above.

the focal user leaves on other users’ content; self-directed accepts are excluded. Within each window we track when (and whether) each such event occurs, and the Cox model estimates the instantaneous hazard of helping as a function of time-varying covariates that capture the phase of the observation timeline and the treatment status. We report the answers-only and comments-only outcomes separately in the decomposition (Table 6).

Why answers anchor the construct.. Broadening the outcome to include comments responds to the concern that an answers-only measure is too narrow. We nonetheless treat the posting of an answer as the theoretically central helping act. Generalized reciprocity, as modeled in the experimental literature, is a *costly* prosocial contribution: individuals pass a benefit to an unrelated third party at real expense to themselves (Greiner and Levati, 2005; Stanca, 2009; Mujic and Leibbrandt, 2018), and it is the willingness to bear that cost — not mere approval or endorsement — that distinguishes reciprocity from low-cost signaling (Molm et al., 2007). Answering a stranger’s question is the platform analogue: it demands substantive effort and yields a public good for the community. Lower-cost gestures such as upvotes and edits index curation or approval more than the effortful contribution the construct denotes. We therefore anchor the outcome on answers, add comments as a robustness broadening, and — because the lowest-cost channels (upvotes, edits) lie outside our processed data — read our estimates as a lower bound on total reciprocal behavior (see Section 6.5).

Treatment.. The treatment is the receipt of an answer. We operationalize this using *treatment*, a binary variable indicating whether the user’s question received at least one answer with a non-negative score. We adopt this operationalization rather than Stack Overflow’s “accepted answer” mechanism to address endogeneity: marking an answer as accepted requires the user to return to the site, which may itself correlate with helping behavior.

Response Time.. For each treated observation, the response time variable used in the interaction terms is $\log(1 + t_{\text{answer}})$, where t_{answer} is the elapsed time in hours from the start of the observation window ($T_Q - 2D$) to the arrival of the first answer with a non-negative vote score. Because $T_Q - 2D$ is a fixed origin common to all observations, this is a monotone transformation of the actual response time in hours ($t_{\text{answer}} - t_{\text{question}}$) and preserves the full ordering of fast versus slow answers; however, it shifts the scale by the constant pre-question window length. For numerical stability,

both response-time interaction terms ($\text{isTreatedActive} \times \log(1 + t_{\text{answer}})$ and $\text{treatedPostQuestion} \times \log(1 + t_{\text{answer}})$) are clipped to their 5th–95th percentile range and then standardized (z-scored) before model fitting. Consequently, the reported interaction coefficients γ and δ are expressed per one standard deviation of the respective interaction term rather than per unit of $\log(1 + t_{\text{answer}})$.

User Experience. To test Hypothesis 2, we stratify our analysis by user tenure, defined as the number of days between a user’s first activity on the platform and the start of the observation window. We define seven tenure buckets using right-closed intervals: $[0, 7]$, $(7, 30]$, $(30, 180]$, $(180, 365]$, $(365, 1095]$, $(1095, 2190]$, and $(2190, \infty)$ days, corresponding to the labels less than one week, one week to one month, one to six months, six to twelve months, one to three years, three to six years, and more than six years, respectively. Running separate models by bucket enables us to examine how the reciprocity effect varies across the user trajectory without imposing a parametric functional form.

4.4. Propensity Score Matching

Because the receipt of an answer (the treatment) is correlated with user characteristics (more experienced users, for instance, might tend to ask higher-quality questions that are more likely to receive answers), we employ propensity score matching (Rosenbaum and Rubin, 1983) to improve the comparability of the treatment and control groups. Importantly, matching is performed *across different users*: each treated question (from a user who received an answer) is matched to a control question (from a different user whose question did not receive an answer) that has a similar propensity to receive an answer based on observable pre-treatment characteristics. This cross-user matching ensures that treated and control observations are independent conditional on the matched covariates, while the DiD structure within each pair absorbs residual differences.

We estimate the propensity score using the following pre-treatment covariates, which jointly capture user experience, recent activity, and question characteristics as depicted in Table 2. It should be noted that *tagAnswerRateAvg* is critical for addressing the concern that question complexity may confound the treatment: questions in tags with low answer rates are inherently more difficult, and matching on this rate ensures that treated and control questions come from similarly tractable topic areas. Matching on the

Table 2: Definition of Matching Covariates

Variable name	Definition
<i>Exact matching</i>	
calendarYear	Calendar year exact matching indicating the year of the observation, controlling for secular trends in platform activity and answer rates.
topLevelTag	First tag of the question indicating the main topic and community that sees the question
<i>Propensity score matching</i>	
userTenure	Number of days since the user’s first recorded activity on the platform, capturing cumulative experience.
numQuestionsAskedAT	Total number of questions asked by the user prior to the observation window.
numHelpProvidedAT	Total number of answers provided by the user prior to the observation window.
numQuestionsAsked30D	Number of questions asked in the 30 days preceding the observation window.
numHelpProvided30D	Number of answers provided in the 30 days preceding the observation window.
numQuestionsAsked7D	Number of questions asked in the 7 days preceding the observation window.
numHelpProvided7D	Number of answers provided in the 7 days preceding the observation window.
tagAnswerRateAvg	Average share of questions that receive an answer within the question’s tags, capturing topic-specific difficulty.

tag-level answer rate is particularly important for our response time analysis, as it mitigates the concern that slower response times simply proxy for differently sized sub-communities with lower throughput. We use nearest-neighbor matching with a caliper of 0.05 on the propensity score. Because the pool of unanswered questions is smaller than the pool of answered questions, some control questions are matched to more than one treated question; the matched file therefore contains more rows than unique question pairs (12.3 million pairs, 24.5 million rows). Each row enters the Cox analysis as an independent observation window; standard errors are discussed in the robustness appendix.

Table 3: Covariate Balance Before and After Matching

Covariate	Unmatched			Matched		
	Tr Mean	Ct Mean	SMD	Tr Mean	Ct Mean	SMD
userTenure	570.95	720.62	−0.167	569.58	568.20	0.002
numQuestionsAskedAT	31.11	26.00	0.059	30.75	30.19	0.006
numHelpProvidedAT	15.27	14.15	0.009	14.70	12.87	0.018
numQuestionsAsked30D	2.00	1.19	0.197	1.98	1.96	0.004
numHelpProvided30D	0.81	0.46	0.068	0.78	0.63	0.029
numQuestionsAsked7D	0.57	0.34	0.174	0.56	0.56	−0.001
numHelpProvided7D	0.22	0.12	0.063	0.21	0.17	0.028
tagAnswerRateAvg	0.51	0.44	0.695	0.51	0.51	0.004

Note. SMD = standardized mean difference. The conventional threshold for acceptable balance is $|\text{SMD}| < 0.1$. All post-matching SMDs fall well below this threshold.

Table 3 displays the covariate balance before and after matching. Before matching, several covariates showed a meaningful imbalance between the treated and control groups. For instance, treated users had asked more questions in the prior 30 days ($\text{SMD} = 0.197$) and 7 days ($\text{SMD} = 0.174$), reflecting that users who ask questions that attract answers tend to be more recently active. The largest imbalance was in the tag-level answer rate ($\text{SMD} = 0.695$), confirming that question topic characteristics strongly predict answer receipt and underscoring the importance of matching on this variable. After nearest-neighbor propensity score matching, all standardized mean differences fell well below the conventional threshold of 0.1, with the largest residual imbalance at 0.029 (see Table 3). This confirms adequate balance across all matching covariates, and the balance is also illustrated in the Love plot in Appendix Appendix B.

4.5. Analytical Strategy: Cox Proportional Hazards with Time-Varying Covariates

We model the hazard of helping using a Cox proportional hazards model with time-varying covariates (Andersen and Gill, 1982). The counting-process formulation of Andersen and Gill (1982) extends the original Cox model ((Cox, 1972) to settings with time-varying covariates and recurrent events, providing the large-sample theoretical foundation (including consistency and asymptotic normality of the partial likelihood estimator) that justifies our approach.

For each matched pair of observations (one treated, one control), we partition the 4-day observation window into intervals defined by the key event times: the start of the window, the posting of the question, and the arrival of the answer (the actual answer arrival for treated users or the synthetic transition for controls).

The hazard function is specified as:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\beta_0 \cdot \text{treatment} + \beta_1 \cdot \text{phasePostQuestion}(t) + \beta_2 \cdot \text{treatedPostQuestion}(t) + \beta_3 \cdot \text{phasePostAnswer}(t) + \beta_4 \cdot \text{isTreatedActive}(t) \right) \quad (1)$$

where $h_0(t)$ is the baseline hazard. *treatment* is a time-constant indicator for treatment group membership (1 if the user’s question received an answer, 0 otherwise); it absorbs stable between-group differences in baseline helping rates. The remaining covariates are time-varying indicators that switch on at the appropriate event times. Because *treatedPostQuestion* switches on when the question is posted and remains on through the post-answer phase, the post-answer treatment effect is not the *isTreatedActive* coefficient in isolation but the sum $\beta_2 + \beta_4$ — the difference, between treated and control users, in the change in helping hazard from the pre-question baseline to the post-answer phase. We report this summed difference-in-differences contrast as the treatment effect, together with its two components: β_2 , the waiting-period divergence, and β_4 , the increment at answer arrival net of that divergence.

To test the moderation by response time (Hypothesis 3), we extend the model with two additional terms:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\boldsymbol{\beta}' \mathbf{X}(t) + \gamma \cdot \text{isTreatedActive}(t) \times \widetilde{\log}(\text{RT}) + \delta \cdot \text{treatedPostQuestion}(t) \times \widetilde{\log}(\text{RT}) \right) \quad (2)$$

where $\widetilde{\log}(\text{RT})$ denotes the standardized log response-time variable (see below), γ captures the moderation of the answer-arrival increment by answer speed, and δ captures the analogous moderation during the waiting period. Just as the base treatment effect is the sum $\beta_2 + \beta_4$, the response-time moderation of that effect is the sum $\gamma + \delta$; we report this net moderation alongside its two components. A negative $\gamma + \delta$ would indicate that faster answers

strengthen the reciprocity effect (consistent with H3), while a positive sum would indicate the opposite.

Stratification by Tenure.. To test Hypothesis 2, we fit separate models for each of the seven tenure buckets. This allows the baseline hazard and all coefficients to vary freely across levels of user experience, providing a non-parametric view of how reciprocity changes over the user trajectory.

Identification.. Our identification strategy rests on the matched DiD logic embedded in the model. Propensity score matching (Rosenbaum and Rubin, 1983) ensures that treated and control observations come from users with comparable baseline characteristics, including tenure, activity levels, and question topic difficulty. This matching absorbs between-user differences that might otherwise confound the treatment effect. We do not include user fixed effects in the survival models. Because matching is performed across different users (not within-user), and because the Cox model already leverages within-observation temporal variation through the time-varying covariate structure (Andersen and Gill, 1982), user fixed effects are not required for identification. The DiD structure compares within-observation changes before and after the answer across treatment groups, isolating the treatment effect from stable individual differences. The waiting-period covariate (*treatedPostQuestion*) captures any lift in helping that treated users show after asking but before receiving an answer. When this term is near zero, treated and control users move in parallel up to the answer, and the summed post-answer effect can be read as the effect of answer receipt rather than of the act of asking itself; when it is large, the two groups diverge before treatment arrives, and the summed effect blends this pre-answer divergence with any response to the answer itself. Even a near-zero waiting-period term, however, rules out only motivational shifts that materialize *before* the answer: question-induced channels that operate mainly after the answer arrives — most notably the elevated profile visibility a question generates — are not absorbed by it (see Section 6.5). We therefore report the waiting-period term explicitly, so that departures from parallel trends are measured rather than assumed away, and we return to its magnitude in the Results.

5. Results

5.1. Descriptive Statistics

The Cox analytic sample spans 24,527,534 question-level observations from 3,899,114 unique users on Stack Overflow, with 50% of questions receiving at least one answer. Help-event counts are right-skewed: the mean is 0.31 events per observation window ($SD = 2.27$; median 0), consistent with helping being a relatively rare behavior for any individual user in any given window. Among treated users, the median response time from question to first answer is 0.34 hours, though the distribution is heavily right-skewed (mean 5.09 hours, $SD = 17.09$), reflecting a minority of questions that wait many hours or days for a response. The sample spans all levels of platform experience, with the largest strata being established users (1–3 years: $n = 6,471,198$) and newcomers (< 1 week: $n = 4,049,574$).

Table 4: Descriptive Statistics

Variable	Mean	SD	Median	N
Questions (observations)				24,527,534
Unique users				3,899,114
% questions with answer	50.00%			
Help events per window	0.38	3.21	0.00	
User tenure (days)	663	849	329	
Response time (hours, treated)	5.09	17.09	0.34	
<i>Response time (hours, treated) by Tenure Bucket</i>				
< 1 Week	4.52	15.75	0.32	2,024,689
1 Week - 1 Month	3.60	13.86	0.26	781,028
1 - 6 Months	3.84	14.46	0.26	2,038,312
6 - 12 Months	4.25	15.43	0.28	1,502,751
1 - 3 Years	5.12	17.19	0.34	3,235,547
3 - 6 Years	6.71	19.90	0.50	1,820,870
> 6 Years	8.69	22.76	0.76	860,231
<i>N (unique questions) by Tenure Bucket</i>				
< 1 Week				4,049,574
1 Week - 1 Month				1,562,144
1 - 6 Months				4,076,810
6 - 12 Months				3,005,576
1 - 3 Years				6,471,198
3 - 6 Years				3,641,770
> 6 Years				1,720,462

5.2. Descriptive Evidence: Help Rates and the Parallel Trends Assumption

Figure 2 plots the normalized help rate for treated and control users over the two days before and after question posting. Before the question ($t < T_Q$), treated and control help rates track each other closely, consistent with the parallel-trends assumption and with the effectiveness of propensity-score matching. After the question is posted, treated users' help rate rises modestly relative to controls; the descriptive pattern is consistent with a small newcomer-concentrated association rather than a large platform-wide divergence.

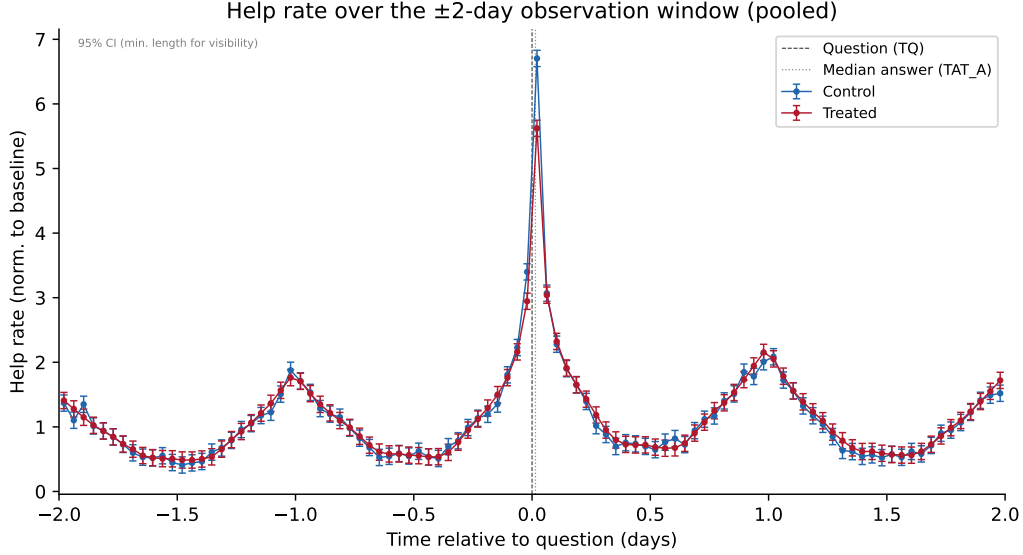


Figure 2: Help rate over the ± 4 -day observation window, pooled across all tenure buckets. The normalized help rate (relative to each group’s pre-question baseline) is plotted for treated users (red, who received an answer) and matched control users (blue, who did not).

5.3. Generalized Reciprocity Effect (H1)

To test Hypothesis 1, we estimate the pooled Cox model across all experience levels (Table 5). The treatment effect — the summed post- versus pre-question contrast ($\beta_2 + \beta_4$) — is positive, small, and precisely estimated: relative to matched controls, treated users’ post-answer helping hazard is about 3% above their pre-question baseline (HR = 1.03, 95% CI [1.02, 1.04], $p < .001$). Hypothesis 1 therefore receives support at the pooled level, but the magnitude is slight, and we show below that it does not survive conditioning on answer quality (Table 10).

The decomposition locates this effect at the arrival of the answer rather than in the waiting period. The waiting-period coefficient is essentially zero and statistically indistinguishable from it ($\beta_2 \approx +0.006$, n.s.), while the answer-arrival increment is small and positive ($\beta_4 \approx +0.023$, $p < .001$); the two sum to the pooled effect above. A waiting-period term this close to zero indicates that the act of asking, by itself, did not shift treated users’ helping during the wait; it does not rule out question-induced influences — such as the increased profile visibility that a question generates — that persist into

the post-answer window (see Section 6.5). We turn next to the answers-versus-comments split.

Table 5: Pooled Cox Regressions (All Experience Levels)

	Main
<i>Treatment Effect (DiD): post-answer vs. pre-question</i>	
Received Answer (net post-answer effect)	0.0492*** (0.0045)
Hazard Ratio [95% CI]	[1.04, 1.06]
Hazard Ratio [bootstrap 95% CI]	[1.02, 1.04]
<i>Decomposition (nested time-varying terms)</i>	
Waiting period: Received Answer $\times$ Post-Question	0.0064
Answer arrival: $\times$ Post-Answer Received	0.0428*** (0.0066)
<i>Response Time Interaction</i>	
Net: Treatment $\times$ log(RT), summed	— —
Component: Post-Answer $\times$ log(RT)	— —
N	24,527,534
Events	4,273,912

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table E.13. Events = helping events in the full analysis sample (after time rounding); when interval rows exceed the fit cap, es
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $^{\dagger}p < 0.1$

Table 6 decomposes the composite outcome by help type. The reciprocity signal is concentrated in *answers*: restricting the outcome to answers the focal user posts to others yields a summed DiD hazard ratio of 1.06 (95% CI [1.05, 1.07]), larger than the answers-plus-comments composite (HR = 1.03, [1.02, 1.04]) that we adopt as the default outcome, whereas comments alone show only a marginal elevation (HR = 1.01, [1.00, 1.03]). In every column the waiting-period component is near zero and the effect enters at answer arrival, mirroring the pooled decomposition. Two implications follow. First, answering — the most effortful and theoretically central form of reciprocal contribution — carries the effect, so the broader composite is, if anything, a conservative reading of the answer-specific signal. Second, adding low-effort comments dilutes rather than amplifies the estimate, which speaks against the concern that an answers-only outcome would mechanically understate newcomer reciprocity by omitting low-cost channels.

Table 6: Reciprocity Effect Decomposed by Help Type (ISS-04)

Outcome	Treatment HR [95% CI]	Waiting β_2	Arrival β_4
Answers only	1.06 [1.05, 1.07]	+0.01	+0.05
Comments only	1.01 [1.00, 1.03]	+0.01	+0.00
Answers + comments	1.03 [1.02, 1.04]	+0.01	+0.02
All types ^a	1.03 [1.02, 1.04]	+0.01	+0.02

Treatment HR is the summed DiD, $\exp(\beta_2 + \beta_4)$; β_2 is the waiting-period (anticipatory-engagement) term and β_4

^a Accept events are treated-only by construction (a control never receives an answer to accept); rows including the Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when genera

5.4. Experience Moderation of Reciprocity (H2)

To test Hypothesis 2, we estimate the model separately within each of the seven tenure buckets (Table 7; Figure 3). The summed treatment effect declines monotonically with tenure. It is largest for the newest users — in the first week, treated users help at an approximately 13% higher post-answer hazard (HR = 1.13, 95% CI [1.12, 1.15], $p < .001$) — remains positive and statistically significant through one to three years (hazard ratios between 1.02 and 1.04), and is indistinguishable from null among users with more than three years on the platform (3–6 years: HR = 1.00, [0.99, 1.01]; over six years: HR = 1.00, [0.98, 1.01]). This graded pattern offers clear support for Hypothesis 2: the reciprocity-like elevation is concentrated at the start of the user trajectory and fades as experience accumulates.

For newcomers, the decomposition is positive in both components: the waiting-period ($\beta_2 \approx +0.056$) and answer-arrival ($\beta_4 \approx +0.070$) terms are each positive and significant and sum to the 13% net effect, whereas among veterans both components are near zero. That newcomers show a positive waiting-period term means part of their elevation begins before the answer arrives — a pattern we return to when discussing the exposure and selection channels the design cannot fully separate from reciprocity (Section 6.5).

The tenure gradient is not an artifact of pooling structurally different entry cohorts. Because calendar year enters the design only as an exact-matching stratum, treated and control questions within a pair always come from the same year; to probe heterogeneity *across* years we re-estimate the pooled effect on the pre-2020 subsample, before the generative-AI disruptions of 2022–2023. The estimate is essentially unchanged (HR = 1.02 pre-2020 versus 1.03 in the full sample; Appendix Table F.14), indicating that the association was already present in the pre-generative-AI platform rather than driven by recent cohorts.

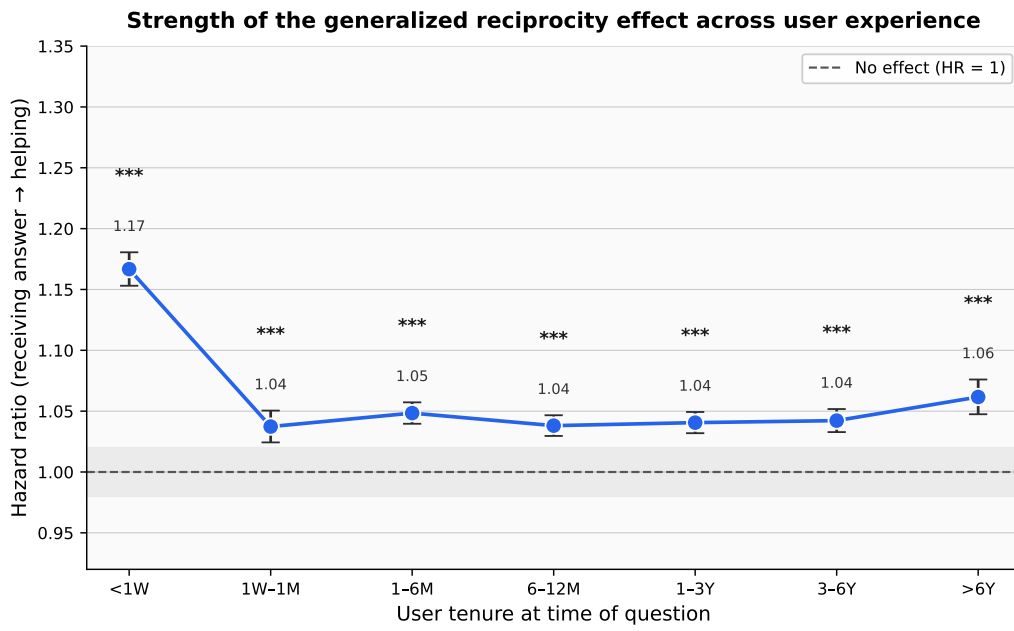


Figure 3: Strength of the generalized reciprocity effect across user experience. Bars show the hazard ratio for the treatment effect by tenure bucket. Error bars indicate 95% confidence intervals.

Table 7: Cox Regression Results: Effect of Receiving an Answer on Helping Hazard

	< 1 Week	1 Wk- 1 Mo	1-6 Mo	6-12 Mo	1-3 Yr	3-6 Yr	> 6 Yrs
<i>Treatment Effect (DiD): post-answer vs. pre-question</i>							
Received Answer (net post-answer)	0.1542***	0.0366***	0.0473***	0.0374***	0.0398***	0.0414***	0.0598***
	(0.0060)	(0.0064)	(0.0043)	(0.0042)	(0.0043)	(0.0046)	(0.0069)
<i>Hazard Ratio</i> <i>[95% CI]</i>	[1.15, 1.18]	[1.02, 1.05]	[1.04, 1.06]	[1.03, 1.05]	[1.03, 1.05]	[1.03, 1.05]	[1.05, 1.08]
<i>Decomposition (nested time-varying terms)</i>							
Waiting period (× Post-Question)	0.0639***	-0.0074	-0.0091	0.0024	0.0084	0.0090	0.0107
Answer arrival (× Post-Answer Received)	0.0903***	0.0440***	0.0563***	0.0350***	0.0313***	0.0324***	0.0492***
N	4,049,574	1,562,144	4,076,810	3,005,576	6,471,198	3,641,770	1,720,462
Events	405,985	347,325	802,364	864,540	837,087	704,702	311,909

N = unique questions (treated + control) in the Cox sample. Within each column, treated vs. control counts can differ because tenure is defined per question.

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table E.13.

Events = helping events in the full analysis sample (after time rounding); when interval rows exceed the fit cap, estimation uses a matched-pair subsample but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $p < 0.1$

5.5. Response Time Moderation (H3)

Hypothesis 3 predicted that faster responses would strengthen the reciprocity effect. Table 8 reports the summed DiD and its response-time moderation, estimated separately by tenure bucket. The net moderation of the treatment effect is the sum $\gamma + \delta$, just as the base effect is $\beta_2 + \beta_4$. The two ways of summarizing the response-time pattern point in opposite directions. The continuous log-response-time interaction is weakly negative for the younger buckets (-0.006 for newcomers, $p < .05$; -0.032 for one week to one month, $p < .001$) and null for the oldest, which taken alone would suggest faster answers slightly strengthen the effect. The more flexible discrete-bin estimates (Table 9, Figure 4) show the reverse, and do so monotonically: the summed DiD rises from essentially null for answers within fifteen minutes ($HR = 0.99$) to 1.11 at twelve to twenty-four hours and 1.30 beyond three days. There is thus no evidence for the immediacy effect H3 predicts — if anything, the treatment effect is *larger* when answers arrive later — and no sign of an intermediate “peak.” We therefore find no support for Hypothesis 3. Because this same speed specification is the one whose base effect falls below one once answer quality is controlled (Table 10), we treat the response-time results as exploratory and do not build interpretation on them.

Table 8: Response Time Moderation of the Reciprocity Effect

	< 1 Week	1 Wk- 1 Mo	1-6 Mo	6-12 Mo	1-3 Yr	3-6 Yr	> 6 Yrs
<i>Base Treatment Effect (DiD, at mean response time)</i>							
Received Answer (net post-answer)	0.0703*** (0.0060)	0.1015*** (0.0069)	0.1263*** (0.0046)	0.1203*** (0.0045)	0.1066*** (0.0046)	0.0835*** (0.0049)	0.0934*** (0.0072)
<i>Net Response-Time Moderation of DiD (summed)</i>							
Treatment $\times$ log(RT), summed	-0.0011 (0.0028)	-0.0275*** (0.0031)	-0.0284*** (0.0020)	-0.0237*** (0.0019)	-0.0110*** (0.0019)	0.0024 (0.0020)	0.0031 (0.0029)
<i>Components</i>							
Answer arrival $\times$ log(RT)	-0.0318*** (0.0023)	-0.0595*** (0.0027)	-0.0641*** (0.0018)	-0.0609*** (0.0017)	-0.0451*** (0.0017)	-0.0219*** (0.0018)	-0.0061* (0.0025)
Waiting period $\times$ log(RT)	0.0307***	0.0320***	0.0357***	0.0372***	0.0341***	0.0243***	0.0093**
N	4,049,574	1,562,144	4,076,810	3,005,576	6,471,198	3,641,770	1,720,462
Events	405,985	347,325	802,364	864,540	837,087	704,702	311,909

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled DiD are in Table E.13.

Events = helping events in the full analysis sample (after time rounding); when interval rows exceed the fit cap, estimation uses a matched-pair subsample but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $^{\dagger}p < 0.1$

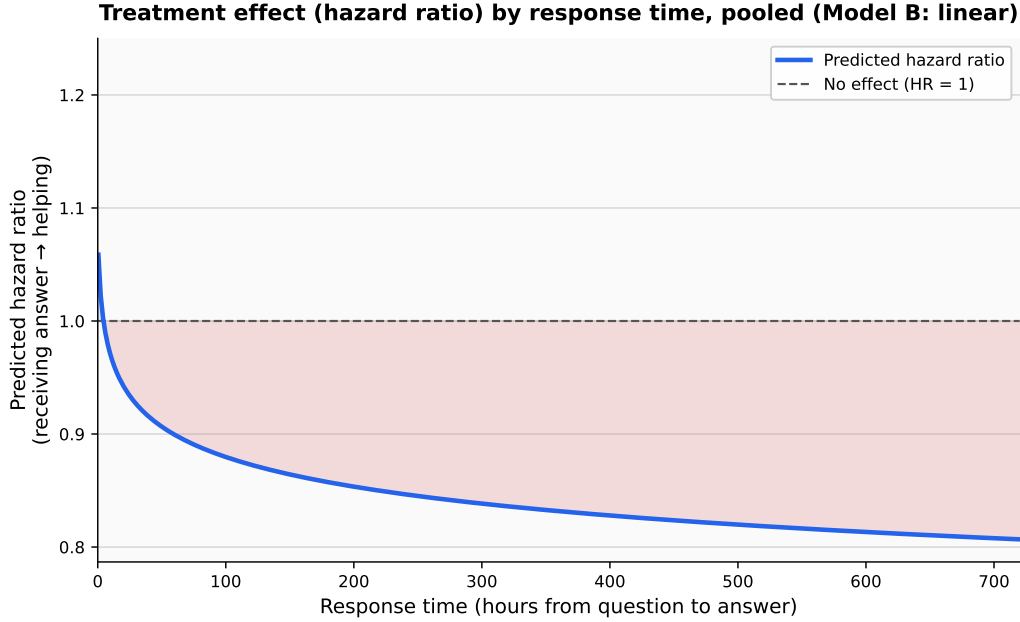


Figure 4: Treatment effect (hazard ratio) by response time bin, pooled across all tenure buckets. Each point represents the estimated hazard ratio of the treatment indicator within that response-time bin, relative to the matched control. Error bars indicate 95% confidence intervals.

Table 9: Treatment Effect by Response-Time Bin

Response time	HR	95% CI	N	E
0-15 min	0.99*	[0.98, 1.00]	17,536,008	66
15-30 min	0.99	[0.98, 1.00]	14,034,514	60
30-60 min	1.02*	[1.00, 1.03]	13,647,991	57
1-2 hr	1.03**	[1.01, 1.05]	13,307,658	57
2-4 hr	1.05***	[1.03, 1.07]	13,054,275	58
4-8 hr	1.06***	[1.04, 1.09]	12,856,408	58
8-12 hr	1.08***	[1.04, 1.11]	12,556,056	58
12-24 hr	1.11***	[1.08, 1.14]	12,739,418	58
1-3 days	1.27***	[1.23, 1.30]	12,686,043	58
>3 days	1.30***	[1.25, 1.35]	12,482,727	59

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated. Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool.

HR is the summed DiD contrast (post-answer vs. pre-question, treated vs. control); comparable across bins.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $^{\dagger}p < 0.1$

Answer quality as an alternative explanation. A natural concern, raised in review, is that response time proxies for answer quality: very fast answers are disproportionately terse or link-only, whereas answers that take longer to arrive tend to be fuller and more detailed, so it may be the quality of the answer — not its timing — that elicits reciprocal helping. We probe this directly by re-estimating the pooled speed specification with controls for three observable quality signals, re-extracted from the raw data dump for this revision: whether the question’s first answer was subsequently accepted, its vote score, and its length in characters (Table 10). The controls do not merely attenuate the effect; they reverse it. The base treatment effect falls from a hazard ratio of 1.10 without quality controls to 0.96 with them, and the same reversal appears within the newcomer stratum, where the quality-controlled hazard ratio is 0.91 (Appendix Table F.15). We read this as a bounding exercise rather than a clean causal decomposition, because the controls are themselves post-treatment and partly endogenous to response speed — acceptance in particular, since marking an answer accepted requires the asker to return to the site, the same re-engagement that produces the helping outcome, so conditioning on it partials out part of the very channel through which reciprocity would operate. With that caveat, the result shows that the modest positive association we estimate does not survive adjustment for answer quality, and that quality is a serious rival to any timing- or gratitude-based reading of the response-time pattern. The one residual is that character count is a coarse proxy for answer quality: it does not distinguish worked code and explanation from boilerplate.

Table 10: Answer-Quality Robustness (ISS-06)

Specification	HR	95% CI	N
speed_baseline	1.095	[1.084, 1.106]	24,527,534
speed_length_only	1.093	[1.082, 1.104]	24,527,534
speed_score_only	1.021	[1.011, 1.031]	24,527,534
speed_quality_controls	0.959	[0.950, 0.968]	24,527,534

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generalizable. Events = helping events in the full analysis sample (after time rounding); when interval rows exceed the fit cap, es

One interpretation of the response-time pattern draws on platform session dynamics: users who post a question may remain actively engaged while waiting, elevating helping during the waiting period regardless of whether an answer arrives. On this account, a fast answer might discharge that

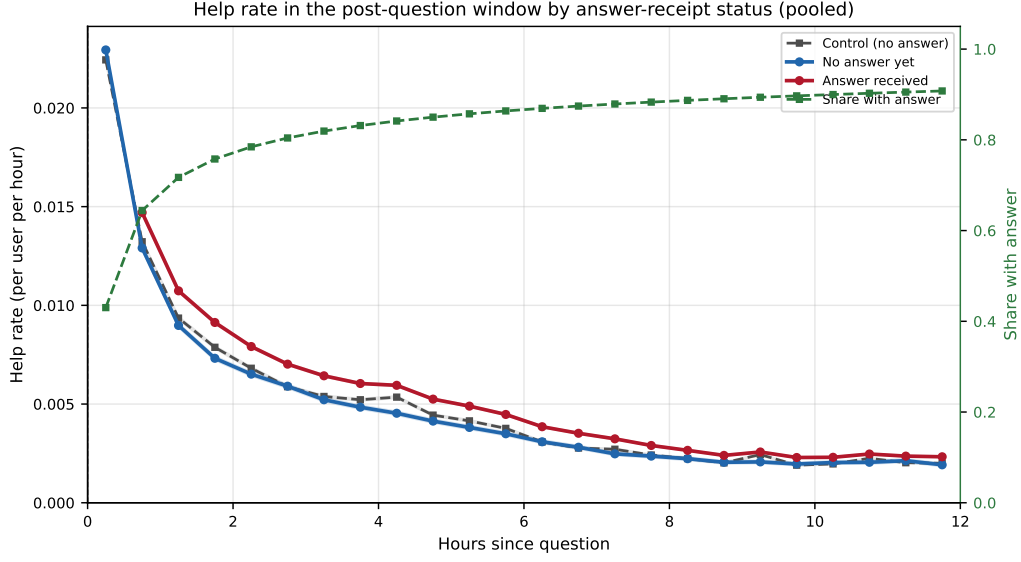


Figure 5: Help rate in the post-question window by answer-receipt status, pooled across all tenure buckets. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour.

pending engagement rather than trigger graded reciprocal gratitude. This interpretation is speculative — we do not observe session boundaries (logins, logouts, inactivity gaps). The answer-quality account above is an equally plausible reading of the same pattern, and a third, purely selective account is that users whose questions attract answers were simply more active to begin with. The design cannot adjudicate among these, and we present the session-dynamics reading as one candidate among several rather than as an established mechanism.

Figure 5 provides descriptive context, comparing users who have received an answer (red) with users who will receive one but are still waiting (blue). Tenure-stratified versions of this figure are in Appendix Figure D.9.

6. Discussion

6.1. Methodological Contribution

A persistent difficulty in studying generalized reciprocity from platform data is that the same user characteristics that predict seeking help also pre-

dict giving it. Active, engaged users are more likely both to post questions and to post answers, so correlations between help received and help given can reflect differences in user engagement rather than any causal response to receiving help. Prior observational studies that report positive associations between help-seeking and subsequent helping behavior without accounting for this selection (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006) may have overstated the role of reciprocity. The matched DiD survival design developed here addresses this confound by comparing changes in helping behavior, before and after question posting, between users whose questions received answers and propensity-score-matched users whose questions did not. Entering the waiting period as an explicit term makes any pre-answer divergence visible rather than silently absorbed into the treatment effect. In the present specification, the pooled contrast is small but positive ($HR \approx 1.03$) and steeply newcomer-concentrated — substantially smaller than many earlier reports, and, as the answer-quality check shows, not robust to controlling for the quality of the answer received, consistent with substantial confounding in prior designs.

6.2. *Reciprocity as a Contributor-Recruitment Mechanism*

The summed treatment effect shows a clear tenure gradient: it is largest among the newest users ($HR = 1.13$) and indistinguishable from null among the most experienced ($HR = 1.00$). We cannot directly observe whether this gradient reflects reciprocity, selection into being answered, the quality of the answers received, or the displacement of general moral motives by community-specific incentives — the design identifies behavioral associations, not motivational shifts. If the gradient does reflect reciprocity rather than these alternatives, it would imply a narrow role for generalized reciprocity: an early association concentrated at the point of entry, before community-specific motivations are established.

As users accumulate experience, community-specific incentives may coincide with — and plausibly displace — the general reciprocal impulse, though we cannot directly observe this motivational shift. This is consistent with theories of norm displacement (Bénabou and Tirole, 2006), which show how the introduction of external incentives can crowd out moral motives. On Stack Overflow, reputation systems and status markers provide salient, community-specific reasons to help that are not available to newcomers. The absence of a detectable association among the most established users is consistent with

this account, though it is equally consistent with selection or low statistical power in that stratum.

This reframing also helps account for the inconsistency of findings in the prior literature. Studies that draw primarily on samples of experienced users, or that aggregate across experience levels dominated by experienced contributors, should find weaker reciprocity effects under this account. Studies that capture behavior early in the user trajectory should find stronger ones. The variation in prior findings may thus reflect not noise or methodological failure, but genuine heterogeneity in the effect across the user population.

6.3. A Possible Session-Dynamics Interpretation of the Response-Time Pattern

The response-time analysis was intended to test the prediction, derived from laboratory studies of gratitude, that faster help produces stronger reciprocity. It does not. The continuous interaction is weak and inconsistent in sign across tenure, and the more flexible discrete-bin estimates run the other way, with the treatment effect rising rather than falling as answers arrive later. We therefore find no support for the immediacy hypothesis in this field setting, and — because the same specification’s base effect reverses once answer quality is controlled — we treat the response-time results as exploratory.

To the extent that response time matters at all, its direction is the opposite of what emotional-immediacy accounts predict: the treatment effect is larger for slower answers. Two non-gratitude readings are consistent with this. The first is answer quality: slower answers tend to be fuller and more thoroughly worked, and, as shown above, controlling for observable quality reverses the base effect. The second is platform session dynamics: an answer that arrives while a user is still actively engaged may discharge the engagement that the act of asking created rather than trigger a reciprocal impulse that scales with speed. We advance both only as conjectures; our data cannot distinguish them from each other or from a simple selection account, because we observe neither answer text nor the session boundaries (logins, logouts, inactivity gaps) that a direct test would require.

The response-time results bear only weakly on the norm-displacement account. The summed treatment effect is near zero among the most experienced users, and the response-time moderation is likewise negligible for that group, which is at least consistent with the mechanism as a whole having faded rather than merely dampened. Given how weak and inconsistent the

response-time moderation is across tenure, however, we do not lean on this pattern as evidence for displacement.

6.4. Implications for Platform Design

Because any reciprocity-like elevation is modest and concentrated among newcomers, the practical leverage platforms can derive from it lies primarily in contributor recruitment rather than in sustaining contributions from established members — and even that recruitment effect is small in absolute terms.

The most direct implication is that platforms seeking to grow their contributor base should prioritize ensuring that new users receive answers to their questions. In absolute terms, the newcomer summed DiD hazard ratio of 1.13 (Table 7) translates to an additional helping-event risk of roughly 1.2 percentage points per observation window, with a number-needed-to-treat of approximately 80: about 80 newcomer questions must receive an answer to yield one additional helping event to others in the window (Table E.12). The pooled effect across all tenure levels, though statistically clear, is far smaller in absolute terms (number-needed-to-treat above 1,300) and does not survive answer-quality controls, so the design implication is not a platform-wide helping multiplier but a targeted — and fragile — newcomer onboarding channel.

Our results do not support a distinct response-time lever for design. The response-time pattern is inconsistent across specifications and, in the discrete bins, runs opposite to the immediacy prediction (Tables 8 and 9), so we do not recommend timing interventions built on it. The better-supported emphasis is answer receipt itself — ensuring that newcomers’ questions are answered at all, and that answer notifications are salient — rather than optimizing for a particular response-time window.

A further implication concerns the increasing prevalence of AI-generated answers on question-and-answer platforms (see also the GAI-era limitation below). If AI assistance substitutes for human responses, the human-to-human interaction that could sustain generalized reciprocity may decline. Given that any residual association is concentrated among newcomers, this substitution could weaken an early onboarding channel through which new users currently receive a modest push into helping from human community members.

6.5. Limitations

Several limitations constrain the conclusions that can be drawn from this analysis.

The most important identification caveat is right-truncation and churn: users who post a question and leave the platform before the fixed post-question window ends are under-observed for the outcome in ways correlated with both treatment and helping. We do not implement a Heckman-type correction; instead, Appendix Table E.11 reports transparent bounds on the contrast under alternative assumptions about zero-event controls. Even under optimistic bounds, the pooled effect remains modest; the newcomer contrast is more sensitive to the churn assumption and should be read with that sensitivity in mind.

Our helping outcome is a composite of answers and comments the focal user posts to others (self-directed accepts excluded); the decomposition (Table 6) shows the reciprocity signal is carried by answers, with comments adding little. Two low-cost reciprocal channels that novices may favor — upvotes cast on, and edits of, others’ posts — are absent from our processed data and would require re-parsing the raw dump to recover. To the extent that newcomers reciprocate disproportionately through these unobserved low-effort acts, our estimates understate their total reciprocal behavior, and the reported hazard ratios should be read as a lower bound on reciprocity rather than a complete accounting of it. We also cannot cleanly separate a reciprocal response to receiving help from selection into being answered.

Beyond identification, the design captures a behavioral association but cannot identify the psychological mechanism producing it. Gratitude, moral obligation, incentive displacement, and platform-session dynamics are all *proposed* interpretations consistent with parts of the pattern; the data do not allow us to distinguish among them, and we do not observe session boundaries, reputation salience, or emotional states. A particularly consequential competing explanation is answer quality. Faster answers are systematically terser than slower ones, and controlling for three observable quality signals — whether the first answer was accepted, its vote score, and its length in characters — reverses the pooled base effect (Table 10); the positive association therefore cannot be attributed to answer receipt net of answer quality. These controls are imperfect: acceptance and score are post-treatment and partly endogenous (accepting requires the asker to return to the site), and character count captures answer length only coarsely, without distinguishing worked

code and explanation from filler. The direction is nonetheless clear, and it counts against a quality-free reading of the effect. Future work combining behavioral records with experience-sampling measures of emotional state or moral reasoning during platform use could help disentangle these pathways.

The identification strategy rests on the assumption that, conditional on the matched covariates and the DiD structure, answer receipt is approximately independent of the potential outcomes for helping behavior. The propensity score matching substantially improves the comparability of treated and control observations, as confirmed by post-matching balance diagnostics. Unobserved time-varying factors, including changes in workload, access to alternative information sources, or exogenous variation in a user’s availability, could nonetheless bias the estimates in ways that the design cannot rule out.

A specific unobserved channel of this kind is exposure. Posting a question raises the asker’s profile visibility: other users who encounter the question may view the asker’s profile and contribution history, and this salience can generate reputation concern or social-obligation pressure that is distinct from gratitude for a received answer yet persists into the post-answer window. The waiting-period covariate (*treatedPostQuestion*) absorbs the baseline effect of the act of asking, but not any residual visibility that continues after the answer arrives, so the design cannot fully separate gratitude for answer receipt from this exposure channel. The threat is to mechanism attribution rather than to the existence of the behavioral pattern: the summed DiD contrast remains a well-defined estimate of the change following answer receipt, but part of any elevation could reflect exposure rather than reciprocity. A natural probe — a placebo test asking whether unanswered questions with *high* view counts show a helping lift in a pseudo-post-answer window — is unfortunately not identified with the archival data available to us: the Stack Overflow data dump records *ViewCount* only as a single cumulative snapshot at the dump date, so a question’s view count conflates its age, topic popularity, and years of subsequent (largely search-engine) traffic with the contemporaneous exposure during the days-long observation window that such a placebo would require. Testing the exposure channel directly would require time-stamped view data or repeated dump snapshots that do not exist for our study period.

Response time may partially proxy for question characteristics not fully captured by the tag-level answer rate, since questions that attract fast responses may differ systematically from those that wait. Future work with random treatment or finer-grained measures of question difficulty, such as

text-based complexity or specificity scores, could provide a cleaner test of the response time mechanism.

The session-dynamics and re-engagement interpretations rest on indirect evidence: discrete-bin patterns, continuous speed interactions, and descriptive help-rate figures. Direct tests would require session-level data, including login and logout timestamps or inactivity-based session boundaries. We treat these accounts as speculative conjectures, not established mechanisms.

Stack Overflow is a large, relatively anonymous, gamified platform built around technical question-answering. The patterns we estimate may not transfer to smaller or relation-based communities (enterprise Slack channels, academic peer review, intimate hobby forums) where social ties and non-status motives differ. We therefore refrain from claiming that generalized reciprocity “likely operates” elsewhere; at most, our estimates describe one platform’s pre-/early-generative-AI era.

The analysis window spans Stack Overflow’s inception through 1 April 2025 and should be read as a *historical snapshot*, not a prescriptive guide for contemporary platform design. Since 2023, LLM-based tools have structurally disrupted Stack Overflow’s ecosystem: active question volume and human-to-human answering have declined sharply. The human-answer reciprocity associations we estimate were formed when community answering was the norm; their relevance to platforms increasingly mediated by AI-generated responses is uncertain. The pattern is not, however, confined to the recent AI-affected period: restricting the sample to the pre-2020 era leaves the pooled effect essentially unchanged (Appendix Table F.14), so what we document characterizes the pre-generative-AI platform rather than a transient artifact of its disruption.

Finally, the analysis establishes that any short-run association declines with experience but does not characterize what replaces it. The theoretical argument implicates community-specific norms and status motives, but testing this account would require integrating behavioral analysis with direct measures of those alternative mechanisms, for instance, through surveys or experimental manipulations of reputation visibility.

7. Conclusion

Generalized reciprocity is widely theorized as a mechanism sustaining cooperation on knowledge-sharing platforms, but observational evidence is limited by a persistent confound: active users are more likely both to seek

help and to give it, regardless of whether their questions are resolved. Our matched difference-in-differences survival design confronts this confound directly. The pooled treatment effect is small but positive (summed DiD $HR = 1.03$), steeply concentrated among newcomers ($HR = 1.13$) and undetectable among veterans. The newcomer association is the most robust pattern in the data — it holds across entry cohorts and predates the generative-AI era — yet even it is modest in absolute terms and does not survive controls for the quality of the answer received. Receiving help may nudge newcomers toward helping others, but our estimates cannot, on their own, separate that story from selection into being answered or from the quality of the help that arrives; distinguishing them remains a task for designs with exogenous variation in whether, and how quickly, help arrives.

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Ethics statement

This study analyzed de-identified, publicly available archival records from Stack Overflow. No interaction with human participants occurred, and no personally identifiable information was collected or analyzed. Ethical approval was not required under applicable institutional policies for secondary analysis of public platform data.

Data availability

Stack Overflow public data dumps were obtained from the Internet Archive (<https://archive.org/details/stackexchange>). Analysis code and step-by-step instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review). Processed analytical datasets used in the matching and survival analyses are available from the corresponding author upon reasonable request.

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Appendix A. Replication materials

Analysis code and instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review).

Appendix B. Matching Diagnostics

Figure B.6 shows that there is sufficient common support between the treatment and the control group. The propensity score distribution between the original treated and control groups is similar. The distribution of the matched control is nearly identical to the treated group.

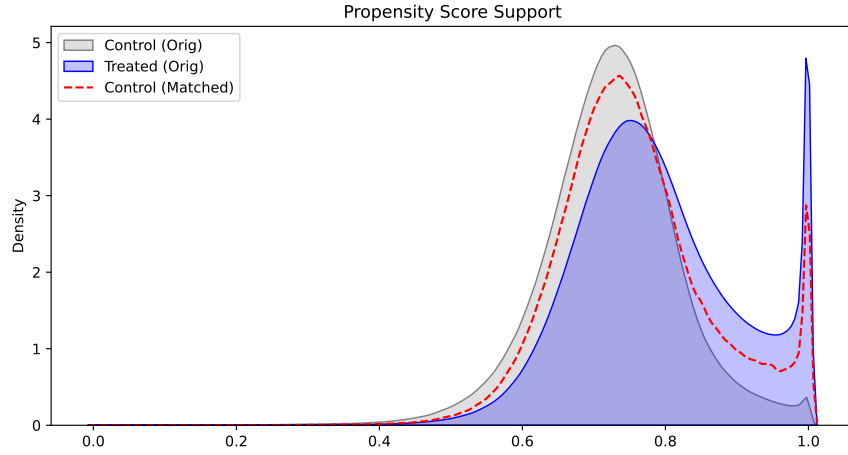


Figure B.6: Distribution of propensity scores of treatment group, control group, and matched control group

Figure B.7 displays the standardized mean differences (SMD) between the matching variables before and after matching. While there are noticeable differences before the matching, all SMDs are very close to zero after the matching.

Appendix C. Help Rate Over the Observation Window by Tenure

Figure C.8 shows that the post-question lift in treated users' help rate is visible across all tenure buckets and is most pronounced among newcomers (tenure < 1 week), consistent with the pattern reported in Table 7 and Figure 3.

Appendix D. Help Rate by Adoption Status and Tenure Bucket

Figure D.9 shows that for experienced users (all panels beyond < 1 Week), the help rate of users still waiting (blue) exceeds that of users who have al-

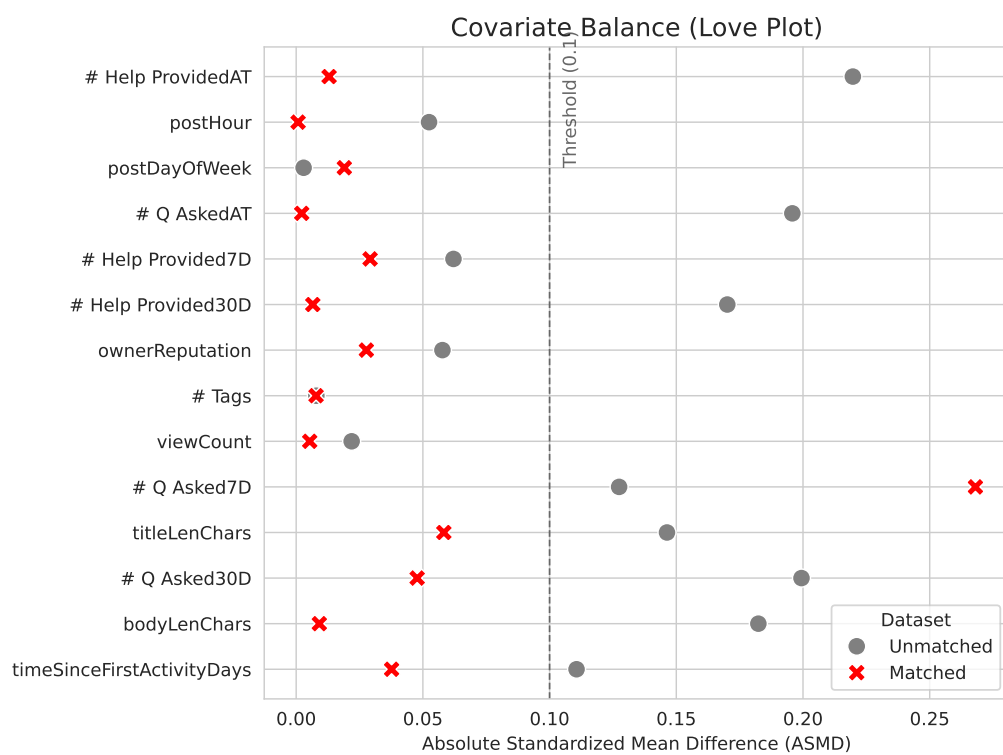


Figure B.7: Visualization of balance in the unmatched and matched dataset

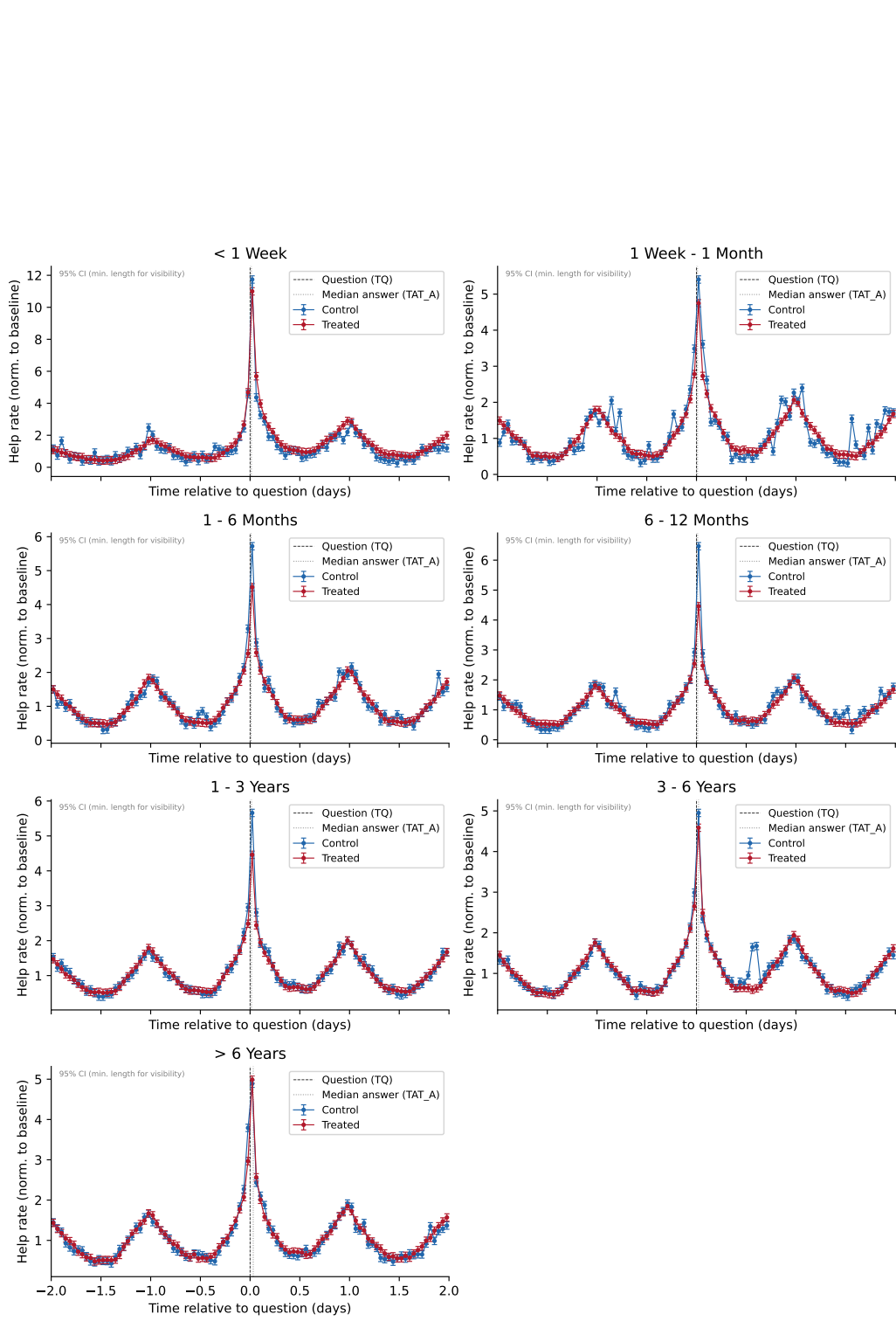


Figure C.8: Help Rate Over the Observation Window, by Tenure Bucket. Each panel plots the normalized help rate (relative to each group’s pre-question baseline) for treated users (red) and matched control users (blue) over the ± 2 -day window around the question-posting time (T_Q , dashed vertical line). The dotted vertical line marks the median answer arrival time (T_A). Error bars indicate 95% confidence intervals.

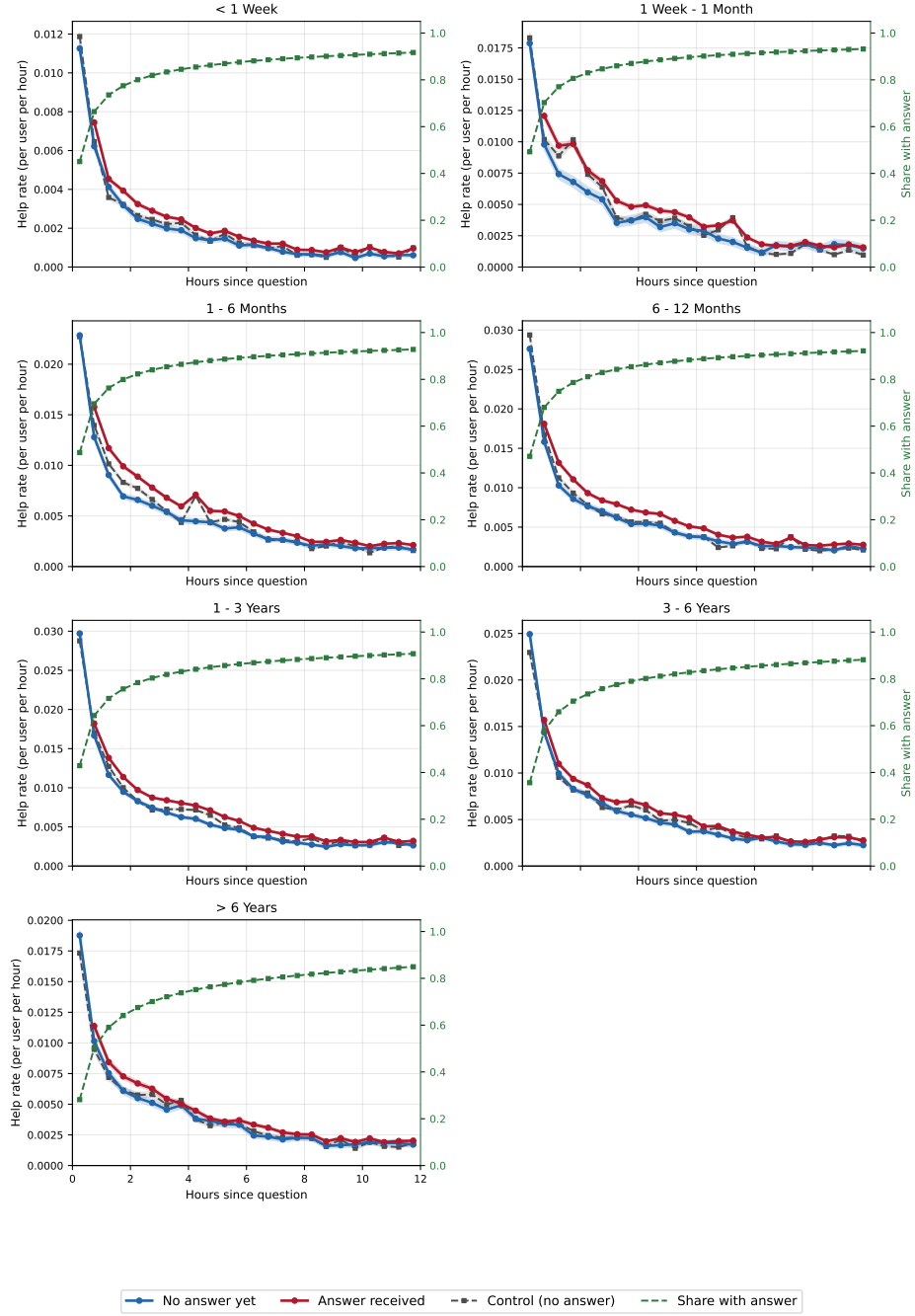


Figure D.9: Help rate in the post-question window by answer-receipt status, separately by tenure bucket.

ready received an answer (red), and both exceed the control baseline (grey dashed), up to approximately 1–2 hours post-question. One reading of this pattern is a session-dynamics account, in which the pending-question state temporarily elevates engagement and answer receipt discharges it; we advance this only as a conjecture, since the design does not observe session boundaries and answer quality offers a competing explanation (Section 6.5). The newcomer panel (< 1 Week) shows the reverse ordering.

Appendix E. Selection Sensitivity and Absolute Effects

Table E.11 reports transparent bounds on the reciprocity contrast under alternative assumptions about zero-event controls. Table E.12 translates the tenure-bucket hazard ratios into absolute-risk-difference proxies using observed baseline event rates. When generated, Table E.13 reports matched-pair bootstrap 95% confidence intervals for the pooled Model A DiD treatment effect (ISS-01).

Table E.11: Selection Sensitivity Based on Zero Post-Question Helping

Scope	Assumed frac.	Base HR	Control zero share	Adjusted RR proxy
All	0.00	1.03	94.4%	1.20
All	0.10	1.03	94.4%	0.75
All	0.25	1.03	94.4%	0.49
All	0.50	1.03	94.4%	0.30
< 1 Week	0.00	1.13	95.1%	0.53
< 1 Week	0.10	1.13	95.1%	0.31
< 1 Week	0.25	1.13	95.1%	0.19
< 1 Week	0.50	1.13	95.1%	0.12
1 Week - 1 Month	0.00	1.02	94.3%	1.03
1 Week - 1 Month	0.10	1.02	94.3%	0.66
1 Week - 1 Month	0.25	1.02	94.3%	0.42
1 Week - 1 Month	0.50	1.02	94.3%	0.27
1 - 6 Months	0.00	1.04	93.8%	1.18
1 - 6 Months	0.10	1.04	93.8%	0.77
1 - 6 Months	0.25	1.04	93.8%	0.51
1 - 6 Months	0.50	1.04	93.8%	0.33
6 - 12 Months	0.00	1.03	93.7%	1.35
6 - 12 Months	0.10	1.03	93.7%	0.89
6 - 12 Months	0.25	1.03	93.7%	0.59
6 - 12 Months	0.50	1.03	93.7%	0.38
1 - 3 Years	0.00	1.02	94.0%	1.41
1 - 3 Years	0.10	1.02	94.0%	0.92
1 - 3 Years	0.25	1.02	94.0%	0.60
1 - 3 Years	0.50	1.02	94.0%	0.38
3 - 6 Years	0.00	1.00	94.8%	1.34
3 - 6 Years	0.10	1.00	94.8%	0.82
3 - 6 Years	0.25	1.00	94.8%	0.52
3 - 6 Years	0.50	1.00	94.8%	0.32
> 6 Years	0.00	1.00	95.9%	1.31
> 6 Years	0.10	1.00	95.9%	0.69
> 6 Years	0.25	1.00	95.9%	0.41
> 6 Years	0.50	1.00	95.9%	0.24

Assumed frac. is the fraction of zero-post-help control questions assigned one latent help event.

Adjusted RR is an event-rate proxy, not a refitted Cox hazard ratio.

Table E.12: Absolute Effect Sizes and NNT

Row	Base risk	HR	ARD	NNT	NNT CI
< 1 Week	0.1003	1.167	0.01671	59.8	55.2–65.2
1 Week - 1 Month	0.2223	1.037	0.00830	120.5	89.1–184.9
1 - 6 Months	0.1968	1.048	0.00953	105.0	88.8–128.2
6 - 12 Months	0.2876	1.038	0.01096	91.3	74.6–117.4
1 - 3 Years	0.1294	1.041	0.00525	190.6	156.7–242.5
3 - 6 Years	0.1935	1.042	0.00817	122.4	99.9–157.6
> 6 Years	0.1813	1.062	0.01117	89.5	72.6–116.2
AllData_Main	0.1742	1.050	0.00879	113.7	95.9–139.4

Table E.13: Matched-Pair Bootstrap Uncertainty for the Summed DiD Treatment Effect (Model A)

Scope	HR	Bootstrap 95% CI	Replicates	N
all	1.03	[1.02, 1.04]	100/100	24,527,534

Replicates resample whole matched pairs with replacement.

Appendix F. Robustness: Cohort Heterogeneity, Observable Selection, and Answer Quality

Table F.14 addresses the concern that pooling entry cohorts from 2008 through 2025 conflates structurally different eras of the platform (reputation inflation, tag evolution, and the post-2022 generative-AI disruption). Because calendar year enters the matching design only as an exact stratum, treated and control questions within a pair always share a year; to probe heterogeneity *across* years we re-estimate the pooled Model A on the pre-2020 subsample. The treatment effect is essentially unchanged ($HR = 1.02$, versus 1.03 in the full sample), indicating that the association is not an artifact of recent cohorts and was already present before the generative-AI era.

Table F.14: Cohort Heterogeneity Robustness (ISS-10)

Specification	HR	95% CI	N	Ev
ModelA_AllData	1.029	[1.020, 1.039]	24,527,534	64
ModelA_Pre2020	1.020	[1.011, 1.029]	19,336,420	73

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generative-AI is excluded.

Table F.15 reports the newcomer (< 1 week) bucket under three specifications: the baseline (HR = 1.14), a specification adding observable pre-treatment selection controls (HR = 1.23), and one adding answer-quality controls (HR = 0.91). The newcomer effect is stable to — indeed slightly amplified by — observable-selection controls, but, like the pooled effect, reverses under answer-quality controls, reinforcing that answer quality is the most consequential confound the design does not resolve.

Table F.15: Newcomer Bucket Robustness (< 1 Week)

Specification	HR	95% CI	N
ModelA_Newcomer_Baseline (< 1 Week)	1.135	[1.121, 1.148]	4,049,574
ModelA_Newcomer_Observable (< 1 Week)	1.226	[1.213, 1.240]	4,049,574
ModelB_Newcomer_Baseline (< 1 Week)	1.054	[1.041, 1.067]	4,049,574
ModelB_Newcomer_LengthOnly (< 1 Week)	1.047	[1.034, 1.059]	4,049,574
ModelB_Newcomer_ScoreOnly (< 1 Week)	0.967	[0.956, 0.979]	4,049,574
ModelB_Newcomer_Quality (< 1 Week)	0.905	[0.894, 0.915]	4,049,574

Cox-table standard errors are model-based; matched-pair bootstrap uncertainty is reported separately when generated. Events = helping events in the full analysis sample (after time rounding); when interval rows exceed the fit cap, es

Appendix G. Analysis of Life-Time Contributions

What remains unclear is whether users who become contributors through generalized reciprocity continue to contribute consistently and become valuable community members. To explore this, we compare the contributions of reciprocity-activated users with those of all contributors selected for Study 1. Specifically, we examine the number of times users from each group provided and received help up to July 14, 2024. The results are presented in Table G.16. Across all quartiles and at the median, reciprocity-activated users ask and provide more help than their counterparts. For example, at the median, reciprocity-activated users provide eight answers and ask 13 questions, compared to six answers and 11 questions among all contributors. Similarly, the 75th percentile of reciprocity-activated users provides 23 answers and asks 29 questions, compared to 19 answers and 22 questions for all contributors. These findings suggest that users who begin contributing after receiving help become equally, if not more, engaged in the community’s helping dynamics compared to all contributors. Furthermore, this analysis demonstrates that once users start contributing, the act of giving and receiving help becomes (at least temporarily) decoupled.

Table G.16: Statistics on Contributions by User Sample

	Answers Provided			Questions Asked		
	Q_1	Med	Q_3	Q_1	Med	Q_3
Reciprocity-activated ($N = 30,160$)	3.0	8.0	23.0	7.0	13.0	29.0
All Contributors ($N = 287,959$)	2.0	6.0	19.0	6.0	11.0	22.0

Appendix H. Implementation Details

All models are estimated using the `lifelines` `CoxTimeVaryingFitter` with a small ridge (L2) penalizer applied for numerical stability. The penalizer strength is $\lambda = 5 \times 10^{-3}$ for main-effect models and $\lambda = 10^{-2}$ for models that include response-time interaction terms (where the interaction covariates are correlated with the base treatment indicator). These values are sufficiently small that they introduce negligible shrinkage relative to the scale of the estimated effects, but they ensure stable Hessian inversion in large risk sets. All covariates are mean-centered before fitting; response-time interaction terms are additionally clipped to their 5th–95th percentile range and z-scored as described above. The standard errors in all tables are model-based (inverse-Hessian) and are *not* corrected for the dependence induced by the matched design — repeated observations per user, reused control questions, and multiple event intervals per timeline. Because that dependence typically makes model-based standard errors too small, the appropriate correction is a cluster-robust (sandwich) variance clustered by matched pair, or a bootstrap that resamples whole matched pairs with replacement; we report matched-pair bootstrap intervals where available and expect them to be somewhat wider than the model-based intervals shown. Ridge penalization and the winsorizing and standardization of the interaction covariates additionally regularize the reported standard errors. For the pooled experienced-user models (H1), estimation was performed on a random sample of approximately 8,001,931 user-interval rows drawn from matched pairs (preserving pair integrity) to keep computation feasible; tenure-bucket-specific models (H2, H3) use all available intervals within each stratum.